

Experiment - 04.

- Aim: Import the data warehouse data in Microsoft Excel and create the Pivot table and Pivot chart.

- Theory:

- (1) Data model is used for building a model where data from various sources can be combined by creating relationships among the data sources. A data model integrates the tables, enabling extensive analysis using Pivot Tables, Power Pivot and power view.

- A data model is created automatically when you import two or more tables simultaneously from a database. The existing database relationships between those tables is used to create the data model in excel.

- (2) The step is as follows -

- Step 1: Open a new blank Workbook in Excel.
- Step 2: Click on the DATA tab.
- Step 3: In the Get External data group, click on the option from Access. The select data source dialog box opens.

- Step 4: Select the accdb file, Event access database file.
- Step 5: The select table window, displaying all the tables found in the database, appears.
- Step 6: Tables in a database are similar to the tables in Excel. Check the 'Enable selection of multiple tables' box. and select all the tables. Then click OK.
- Step 7: The import data window appears. Select the pivot table report option. This option imports the tables into excel and prepares a pivot table for analyzing the imported tables. Notice that the checkbox at the bottom of the window - Add this data to the data model, is selected and disabled.
- Step 8: The data is imported, and a Pivot table is created using the imported tables.

(3). Explore data using pivot table-

Step 1: You know how to add fields to the pivot table and drag fields across areas.

Even if you are not sure of the final report that you want, you can play with the data and choose the best suited report.

In pivot table fields, click on the arrow beside the table - Medals to expand it to show the fields in that table. Drag the NOC- Country Region field in the medals table to the columns area.

Step 2: Drag discipline from the disciplines table to the rows area.

Step 3: Filter discipline to display only five sports: Archery, diving, fencing, figure skating and speed skating. This can be done either Pivot table fields area, or from the row labels filter in the pivot table itself.

Step 4: In pivot table fields, from the medals table, drag medal to the values area.

Step 5: From the medals table, select Medal again and drag it into the filters area.

Step 6: Click the dropdown list button to the right of the Column labels.

Step 7: Select value filters and then select greater than

Step 8: Click OK. The value filters dialog box for the count of Medals is greater than appears.

Step 9: Type 80 in the right field.

Step 10: Click OK.

The pivot table displays only those regions, which has more than total 80 medals.

(4) Create relationship between tables -

Relationships let you analyze your collections of the data in excel, and create interesting and aesthetic reports from the data you import.

Step 1: Insert a new worksheet.

Step 2: Create a new table with new data. Name the new table as what you need.

Step 3: Now you can create relationship between this new table and the other tables that already exist in the data excel model. Rename the sheets as per your preference.

On sheet1, in the pivot table fields list click All. A complete list of available tables will be displayed. The newly added table - will also be displayed.

Step 4: Click on a field in expanded list of fields select sports. Excel messages you to create a relationship between tables.

Step 5: Click on create. The create relationship dialog box opens.

Step 6: To create the relationship, one of the tables must have a column of unique, non-repeated values. In the disciplines table, id column has such values. The table which we have created also has the id column, in table select disciplines.

Step 7: In column (foreign), select sportID. for example.

Step 8: In related table select sports.

Step 9: In related column (primary), id gets selected automatically. Click OK.

Step 10: The pivot table is modified to reflect the addition of the new Data field. Adjust the order of the fields in the rows area to maintain the Hierarchy.

In this case, sport should be first and discipline should be next as discipline will be nested in sports as a sub category.

(5). The analysis helps in -

- Identifying high performance regions.
- Observing monthly sales trends.
- Supporting business decision making.

* CONCLUSION:

Using microsoft Excel Pivot tables and pivot charts, large data warehouse datasets can be easily summarized and visually analyzed.

This method improves -

- Data interpretation.
- Reporting efficiency.
- decision making accuracy.

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